INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

# ME 491
# ENGINEERING ECONOMY & MANAGEMENT

Unit 2

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# Planning

# Planning

- Planning is a decision making process.

- A plan is the route to an objective.

# What is Planning?

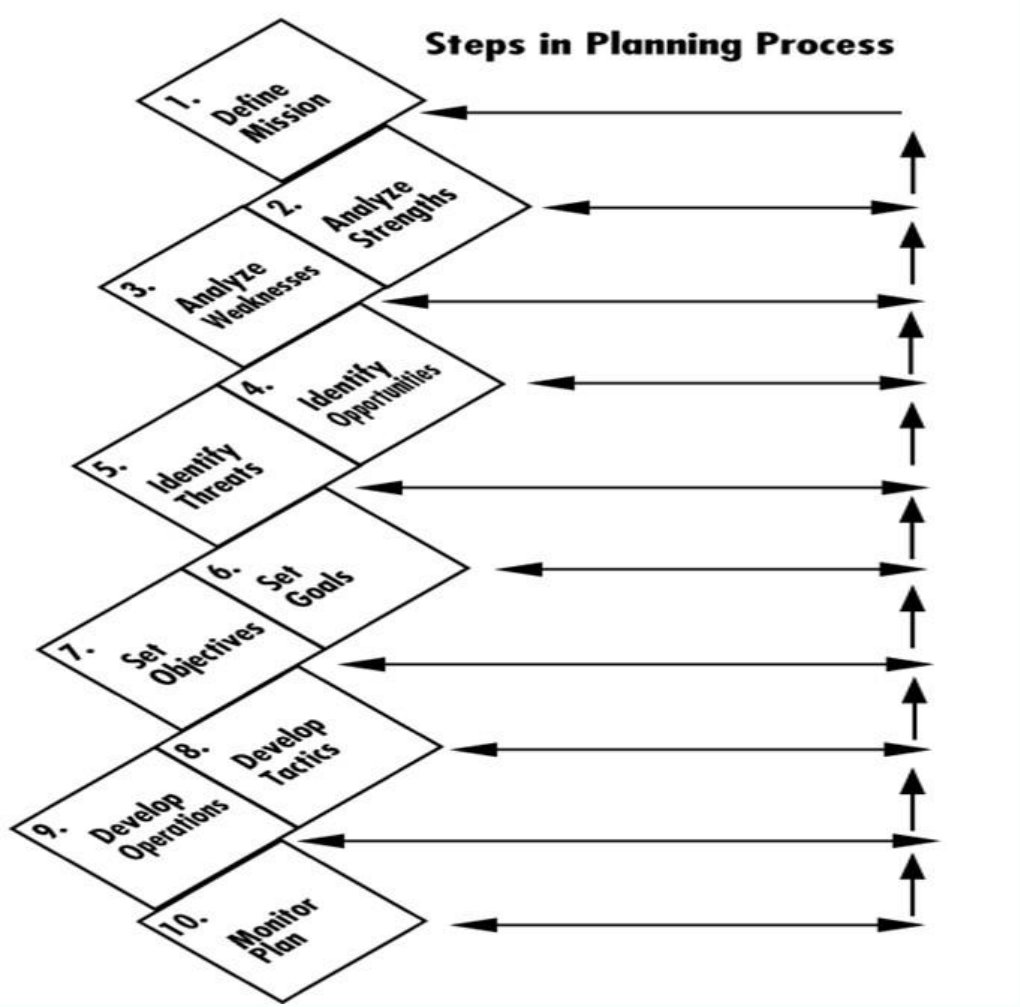
- Planning is the process of deciding what objectives to pursue during a future time period and what to do to achieve those objectives.

- Planning is the first management function.

# What is Planning?

- **Planning is a process that involves defining** the organisation's goals, **establishing** an overall strategy for achieving those goals, and **developing** a comprehensive set of plans to integrate and co-ordinate organisational work.

# The Planning Process



# Planning

- Planning involves the setting of the organisational vision, goals, and objectives.

# Vision

- A vision is more than a goal. It is the larger explanation of why the organisation exists and where it is trying to head.
- A vision gives the organisation a sense of purpose and values to unite workers in a common destiny.
- Managing an organisation without a common vision can be counter productive.

# Vision Statement

- A vision statement is a vivid idealized description of a desired outcome that inspires, energizes and helps you create a mental picture of your target.

- It could be a vision of a part of your life, or the outcome of a project or goal.

# Vision Statement of KNUST

- Is to be globally recognized as the premier centre of excellence in Africa for teaching in Science and Technology for development, producing high calibre graduates with knowledge and expertise to support the industrial and socio-economic development of Ghana and Africa.

# Goals

- Goals are the broad, long term accomplishments an organisation wishes to attain.

- These goals need to be mutually agreed upon by workers and management.

- Thus, goal setting is often a team process.

# Types of goals

- **Strategic goal**: A goal set by and for top management of the organisation.

- **Tactical goal**: A goal set by and for middle managers of the organisation

- **Operational goal**: A goal set by and for lower-level managers of the organisation.

# Plan

- A plan is a blueprint for goal achievement and specifies the necessary resource allocations, schedules, tasks and other actions.

- Goals specify future ends while Plans specify today's ends.

# Plan

- The word planning usually incorporates both ideas (goals & plans).

- Planning means determining the organisation's goals and defining the means for achieving them.

# Objectives

- **Objectives are the specific, short term statements detailing how to achieve the goals.**

- **Objectives** relate to mission statement

- **Explains**
  - **What to do**
  - **How to do it**
  - **When to do it**
  - **Who is to do it**

# Purpose of Planning

- Planning gives direction; reduces the impact of change; minimises waste and redundancy; and sets the standards used in controlling.

## Planning and Performance

- Formal planning is associated with
  - **higher profits,**
  - **higher return on assets, and**
  - **other positive financial results.**

# Planning and Performance

- The quality of the planning process and appropriate implementation of the plans probably contribute more to high performance than does the extent of planning.

# Planning and Performance

- In those studies in which formal planning did not lead to higher performance, the external environment often was the culprit.

- Planning/Performance relationship is influenced by the planning time frame.

## Formal Planning

- Informal planning occurs when planners do not record their thoughts but rather carry them around in their heads.

# Formal Planning

- A formal plan is a written, documented plan developed through an identifiable process.

- The need for formal planning is not limited to large organisations; small ones can realize the same benefits.

# Formal Planning

- Formal planning processes can vary greatly from large to small organisations and even among similar-sized organisations.

- One might have a 2-page formal plan; another 100-page document.

# Formal Planning

- The appropriate degree of sophistication depends on the needs of the individual managers and the organisation itself.

- The environment, size, and type of business are factors that typically affect the planning needs of an organisation.

# Who should plan?

- First line supervisors set objectives on a day-by-day breakdown of what needs to be done.
- Middle level managers prepare plans to aid in the accomplishment of their objectives.
- Top level managers develop plans that guide the entire organisation.

## Who should plan?

- **Formal planning should be practiced by all levels of management - from first-line supervisors right up to the chairman of the board.**

# Functional Plans

- Plans are often classified by function or use.

- The most frequently encountered types of functional plan are

- 1. sales and marketing plans,
- 2. production plans,
- 3. financial plans, and
- 4. personnel plans

# Functional Plans

- Many functional plans are inter-related and inter-dependent.

- For example, a financial plan would be dependent on production, sales, and personnel plans.

## Planning Horizon

- Planning Horizon:
  - Short Range,
  - Intermediate, and
  - Long Range

# Planning Horizon

- Short-range plans generally cover up to one year.
- Long-range plans span at least three to five years, with some extending as far as 20 years into the future.
- Long-range planning is primarily carried out at the top levels of the organisation.

# Planning Horizon

- Intermediate plans cover that time span between short-range and long-range plans.

- Intermediate plans generally cover from one to three or one to five years, depending on the horizon covered by the long-range plan.

# Planning Horizon

- Usually intermediate plans are derived from long-range plans, and short-range plans are derived from intermediate plans.

# Planning Horizon

- For example, if the long-range plan calls for a 40 percent growth in sales by the end of five years, the intermediate plan should outline the necessary steps to be taken over the time span covering one to five years.

- Short-range plans would outline what actions are necessary within the next year.

# Types of Plans

- The most popular ways to describe organisational plans are by their

| | |
|---|---|
| **Breath** | **Strategic, tactical or operational** |
| **Time frame** | **Short term versus long term** |
| **Specificity** | **Directional versus specific, and** |
| **Frequency of use** | **Single use versus standing** |

# Strategic, tactical operational Plans

## Strategic Plans:

- Plans that apply to the entire organisation, establish the organisation's overall goals, and seek to position the organisation in terms of its environment.

# Strategic, tactical operational Plans

- **Tactical Plans**:
- Plans aimed at achieving tactical goals and is developed to implement parts of a strategic plan.

- **Operational Plan**:
- Plans that specify the details of how the overall goals are to be achieved.

# Long term & Short-term Plans; Specificity

- **Long term**: Plans with a time frame beyond three years.
- **Short-term**: Plans covering one year or less.
  - **Specific plans**: Plans that are clearly defined and that leave no room for interpretation.
  - **Directional plans**: Plans that are flexible and that set out general guidelines.

# Frequency of use

- **Single-use plan**: A one time plan specifically designed to meet the needs of a unique situation.

- **Standing plans**: Ongoing plans that provide guidance for activities performed repeatedly.

# Strategic Plans

- Strategic planning is analogous to top level, long-range planning.

- Strategic planning is the planning process applied at the highest levels of the organisation, covering a relatively long period and affecting many parts of the organisation.

# Operational Plans

- Operations or tactical planning is short-range planning and concentrates on the formulation of functional plans.

- Production schedules and day-to-day plans are examples of operational plans.

# Strategic and Operations Planning

- The distinctions between strategic and operations planning are relative, not absolute.
- The major difference is the level at which the planning is done:
- Strategic planning is primarily done by top level managers whereas
- operational planning is done by managers at all levels in the organisation and especially by middle- and lower-level managers.

# Organisational Mission

- An organisation's mission is actually the broadest and highest level of objectives.

- Mission defines the basic purpose or purposes of the organisation (for this reason, the terms mission and purpose are often used interchangeably).

# Organisational Mission

- An organisation's mission outlines why the organisation exists.

- A mission statement usually includes a description of the organisation's basic products and/or services and a definition of its markets and/or sources of revenue.

# Organisational Mission

- There are 3 questions that need to be answered to define an organisation's present business
- 1. Management must identify the customers - where they are, how they buy, and how they can be reached.
- 2. Management must know what the customer buys.
- 3. What is the customer looking for in the product?

# Mission Statement of KNUST

- The mission of KNUST is to provide an environment for teaching, research and entrepreneurship training in science and technology for development in Ghana and Africa.

- KNUST will also provide services to the community, be open to all the people of Ghana and positioned to attract scholars, industrialists and entrepreneurs from Africa and other international communities.

# Management by Objectives (MBO)

- **Management by objectives (MBO)** is a method whereby managers and employees define goals for every department, project, and person and use them to monitor subsequent performance.

# Management by Objectives (MBO)

- "What are we trying to accomplish?"

- **A good goal should be concrete and realistic, provide a specific target and time frame, and assign responsibility.**

# Management by Objectives (MBO)

- Goals may be quantitative or qualitative, depending on whether outcomes are measurable.

# Management by Objectives (MBO)

- **Quantitative goals** are **described in numerical terms** such as 'Salesperson Jones will obtain 16 new accounts in December."

- **Qualitative goals** use statements such as 'Marketing will reduce complain by improving customer service next year."

# Management by Objectives (MBO)

- Goals should be jointly derived.

- **Mutual agreement between employee and supervisor creates the strongest commitment to achieving goals.**

- **In the case of teams, all team members may participate in setting goals.**

## Policies

- A policy is a deliberate plan of action to guide decisions and achieve rational outcome(s).

# Policies

- The term may apply to government, private sector organisations and groups, and individuals.

- Policies are broad, general guides to action that constrain or direct objective attainment.

# Policies

- Policies do not tell organisational members exactly what to do, but they do establish the boundaries within which they must operate.

- For example, a policy of answering all written customer complaints in writing within 10 days" does not tell a manager exactly how to respond, but it does say it must be done in writing within 10 days.

# Policies

- Policies create an understanding among members or group that makes the actions of each member more predictable to other members.

- Policy statements often contain the words 'to ensure", 'to follow', 'to maintain', 'to promote', 'to be', 'to accept', and similar verbs.

# Policies

- For example, a Company XYZ, may have a policy "to accept all returns that are accompanied by a sales slip."

- Such a policy is a general guideline for pursuing company objectives related to profit and sales.

# Procedures and Rules

- **Procedures** and rules differ from policies only in degree. In fact, they may be thought of as low-level policies.

- A **procedure** is a series of related steps or tasks expressed in chronological order for a specific purpose.

# Procedures and Rules

- Procedures define in step-by-step fashion the methods through which policies are achieved.
- Procedures emphasise details.
- They outline precisely how a **recurring activity** must be accomplished.
- 
- Procedures allow for little flexibility and deviation.

# Procedures and Rules

- A company's policy may be to accept all customer returns submitted within one month of purchase;
- company procedures would outline exactly how a return should be processed.
- Well-established, formal procedures are often known as standard operating procedures (SOPs).

# Rules

- Rules require specific and definite actions to be taken or not to be taken in a given situation.
- Rules leave little doubt about what is to be done.
- They permit no flexibility and deviation.
- Unlike procedures, rules do not have to specify sequence. For example, “No smoking in the conference room” is a rule.

# Policies, Procedures and Rules

- Procedures and rules are actually subsets of policies.

- All provide some kind of guidance.

# Policies, Procedures and Rules

- The differences lie in the range of applicability and the degree of flexibility.

- A no-smoking rule is much less flexible than a procedure for handling customer complaints; the latter is likewise less flexible than a hiring policy.

# Tools for Planning and Decision Making

- To plan managers must make assumptions about future events.
- Planners must develop forecasts of probable future circumstances.
- Forecasting is the process of developing assumptions or premises about the future that managers can use in planning or decision making.

# Tools for Planning and Decision Making

**Forecasts: Predictions of outcomes**

- **Types of forecasts:**

- **Two specific outcomes managers attempt to forecast are future revenues and new technological breakthroughs.**

# Tools for Planning and Decision Making

## Revenue forecasting

- Revenue forecasting involves predicting revenues.

- Revenue forecasting is a critical element of planning for both profit and non-profit organisations.

# Tools for Planning and Decision Making

## Technological forecasting

- Technological forecasting involves predicting changes in technology and when new technologies are likely to be economically feasible.

# Tools for Planning and Decision Making

## Forecasting Techniques

- **Forecasting techniques fall into two categories.**

- **1. Quantitative forecasting** applies a set of a series of past data to predict outcomes.

- **2. Qualitative forecasting** uses the judgement and opinions of knowledgeable individuals to predict outcomes.

# Tools for Planning and Decision Making

- **Benchmarking** is the search for the best practices among competitors or non-competitors that lead to their superior performance.

**Steps in Benchmarking**

- 1. Form a benchmarking planning team
- 2. Gather internal and external data
- 3. Analyse data to identify performance gaps
- 4. Prepare data to identify performance gaps
- 5. Prepare and implement action plan
- **------→ BEST PRACTICES!**

# Tools for Planning and Decision Making

## Budgets

- A budget is a numerical plan for allocating resources to specific activities.

## Types of budgets

- 1. Revenue budgets
- 2. Expense budgets
- 3. Cash budgets

# Tools for Planning and Decision Making

- **Scheduling** – A list of necessary activities, their order of accomplishment, who is to do each, and the time needed to complete them.

- **Gantt chart:** A scheduling chart developed by Henry Gantt that shows actual and planned output over a period of time.

# Tools for Planning and Decision Making

- **Breakeven analysis:** A technique for identifying the point at which total revenue is just sufficient to cover total costs.

- **Linear Programming:** A mathematical technique that solves resource allocation problems.

# Tools for Planning and Decision Making

## Causal Modelling

- A group of different techniques that determine causal relationships between different variables.

- ***Regression models***
- ***Econometric models***
- ***Economic indicators***

# Tools for Planning and Decision Making

*Regression models*

- Used to predict one variable (called the dependent variable) on the basis of known or assumed other variables (called independent variables).

- For example, we might predict future sales based on the values of price, advertising, and economic levels.

# Tools for Planning and Decision Making

## *Econometric models*

- Make use of several multiple-regression equations to consider the impact of major economic shifts.

# Tools for Planning and Decision Making

## *Economic indicators*

- Are various population statistics, indexes, or parameters that predict organisationally relevant variables such as discretionary income.

- Examples include cost-of-living index, inflation rate, and level of unemployment.